# Lab 3: Every subagent succeeds, and the report still misses half the topic

Topic 1: Agentic Architecture & Orchestration · Task 1.2: Orchestrate multi-agent systems with coordinator-subagent patterns

Runs on Simulator 3: Multi-Agent Research System · Run in class

Source: Official exam guide, sample question 7

## Commit to an answer first

After running the system on the topic "impact of AI on creative industries," you observe that each subagent completes successfully: the web search agent finds relevant articles, the document analysis agent summarizes papers correctly, and the synthesis agent produces coherent output. However, the final reports cover only visual arts, completely missing music, writing, and film production. When you examine the coordinator's logs, you see it decomposed the topic into three subtasks: "AI in digital art creation," "AI in graphic design," and "AI in photography." What is the most likely root cause?

**A.** The synthesis agent lacks instructions for identifying coverage gaps in the findings it receives from other agents.

**B.** The coordinator agent's task decomposition is too narrow, resulting in subagent assignments that don't cover all relevant domains of the topic.

**C.** The web search agent's queries are not comprehensive enough and need to be expanded to cover more creative industry sectors.

**D.** The document analysis agent is filtering out sources related to non-visual creative industries due to overly restrictive relevance criteria.

## Simulate the solution

```
# In the browser: open Simulator 3, then Lab 3
cd labs
uv run serve.py

# In Claude Code, from the labs folder
/lab 3

# In a terminal, from simulators/simulator-3-multi-agent-research/
uv run research_system.py --narrow
uv run research_system.py
```

## What you should see

The narrow run never searches music, writing or film. The broad run does.

Open `simulators/simulator-3-multi-agent-research/evidence/` and find it yourself.

## The answer

**B.** Decompose broadly: music, writing, film and visual arts.

The coordinator's logs reveal the root cause directly: it decomposed "creative industries" into only visual arts subtasks (digital art, graphic design, photography), completely omitting music, writing, and film. The subagents executed their assigned tasks correctly—the problem is what they were assigned. Options A, C, and D incorrectly blame downstream agents that are working correctly within their assigned scope.

## Now change one thing

In fixtures/sources.json replace the visual-arts TIMEOUT with [] and rerun: a valid empty result must not read as a failure.